

as “near to”), we use a contrastive caption generation approach. Specifically, first, a pair of images, each containing two concepts of interests, would be randomly sampled from MS COCO (we use the train and validation sets of COCO 2017). Second, an annotator would be given a template containing the two concepts and is required to choose a spatial relation from a pre-defined list (Table 1) that makes the caption correct for one image but incorrect for the other image. We will detail these steps and explain the rationales in the following.

Image pair sampling. MS COCO 2017 contains 123,287 images and has labelled the segmentation and classes of 886,284 instances (individual objects). Leveraging the segmentation, we first randomly select two concepts (e.g., “cat” and “laptop” in Fig. 3), then retrieve all images containing the two concepts in COCO 2017 (train and validation sets). Then images that contain multiple instances of any of the concept are filtered out to avoid referencing ambiguity. For the single-instance images, we also filter out any of the images with instance pixel area size $< 30,000$, to prevent extremely small instances. After these filtering steps, we randomly sample a pair in the remaining images. We repeat such a process to obtain a large number of individual image pairs for caption generation.

Fill in the blank: template-based caption generation. Given a pair of images, the annotator needs to come up with a valid caption that makes it a correct description for one image but incorrect for the other. In this way, the annotator should focus on the key difference between the two images (which should be a spatial relation between the two objects of interest) and choose a caption that differentiates the two. Similar paradigms are also used in the annotation of previous vision-language reasoning datasets such as NLVR(2) (Suhr et al., 2017, 2019) and MaRVL (Liu et al., 2021). To regularise annotators from writing modifiers and differentiating the image pair with things beyond accurate spatial relations, we opt for a template-based classification task instead of free-form caption writing.² Besides, the template-generated dataset can be easily categorised based on relations and their categories. Specifically, the annotator would be given instance pairs as shown in Fig. 3.

²Hendricks and Nematzadeh (2021) propose a zero-shot probing benchmark of similar spirit for *verb* understanding. All captions are simplified as subject-verb-object triplets.

The caption template has the format of “*The ENT1 (is) _____ the ENT2.*”, and the annotators are instructed to select a relation from a fixed set to fill in the slot. The copula “is” can be omitted for grammaticality. For example, for “contains” and “has as a part”, “is” should be discarded in the template when extracting the final caption.

The fixed set of spatial relations enable us to obtain the full control of the generation process. The full list of used relations are listed in Table 1. It contains 71 spatial relations and is adapted from the summarised relation table of Marchi Fagundes et al. (2021). We made minor changes to filter out clearly unusable relations, made relation names grammatical under our template, and reduced repeated relations. In our final dataset, 66 out of the 71 available relations are actually included (the other 6 are either not selected by annotators or are selected but the captions did not pass the validation phase).

3.2 Second-round Human Validation

In the second-round validation, every annotated data point is reviewed by at least 3 additional human annotators (validators). Given a data point (consisting of an image and a caption), the validator gives either a `True` or `False` label as shown in Fig. 4 (the original label is hidden). In our final dataset, we exclude instances with fewer than 2 validators agreeing with the original label.

Design choice on reference frames. During validation, a validator needs to decide whether a statement is true or false for an image. However, as discussed in §1, interpreting a spatial relation requires choosing a *frame of reference*. For some images, a statement can be both true and false, depending on the choice. As a concrete example, in Fig. 1, while the potted plant is on the left side from the viewer’s perspective (relative frame), the potted plant is at the right side if the bench is used to define the coordinate system (intrinsic frame).

In order to ensure that annotations are consistent across the dataset, we communicated to the annotators that, for relations such as “left”/“right” and “in front of”/“behind”, they should consider both possible reference frames, and assign the label `True` when a caption is true from either the intrinsic or the relative frame. Only when a caption is incorrect under both reference frames (e.g., if the caption is “*The potted plant is under the bench.*” for Fig. 1) should a `False` label be assigned.

Category	Spatial Relations
Adjacency	Adjacent to, alongside, at the side of, at the right side of, at the left side of, attached to, at the back of, ahead of, against, at the edge of
Directional	Off, past, toward, down, deep down*, up*, away from, along, around, from*, into, to*, across, across from, through, down from
Orientation	Facing, facing away from, parallel to, perpendicular to
Projective	On top of, beneath, beside, behind, left of, right of, under, in front of, below, above, over, in the middle of
Proximity	By, close to, near, far from, far away from
Topological	Connected to, detached from, has as a part, part of, contains, within, at, on, in, with, surrounding, among, consists of, out of, between, inside, outside, touching
Unallocated	Beyond, next to, opposite to, after*, among, enclosed by

Table 1: The 71 available spatial relations. 66 of them appear in our final dataset (* indicates not used).



Caption: *The pizza is at the edge of the dining table.*

The caption is: ☒ True ☐ False

Figure 4: A second-round validation example.

On a practical level, this adds difficulty to the task, since a model cannot naively rely on pixel locations of the objects in the images, but also needs to correctly identify orientations of objects. However, the task is well-defined: a model that can correctly simulate both reference frames would be able to perfectly solve this task.

From a theoretical perspective, by involving more diverse reference frames, we are also demonstrating the complexity of human cognitive processes when understanding a scene, since different people approach a scene with different frames. Attempting to enforce a specific reference frame would be methodologically difficult and result in an unnaturally restricted dataset.

3.3 Annotator Hiring and Organisation

Annotators were hired from prolific.co. We required them to (1) have at least a bachelor’s degree, (2) be fluent in English, and (3) have a >99% historical approval rate on the platform. All anno-

tators were paid 12 GBP per hour.

For caption generation, we released the task with batches of 200 instances and the annotator was required to finish a batch in 80 minutes. An annotator could not take more than one batch per day. In this way we had a diverse set of annotators and could also prevent annotators from becoming fatigued. For second-round validation, we grouped 500 data points in one batch and an annotator was asked to label each batch in 90 minutes.

In total, 24 annotators participated in caption generation and 45 participated in validation. 4 people participated in both phases, which should have minimally impacted the validation quality. The annotators had diverse demographic backgrounds: they were born in 15 countries, were living in 13 countries, and had 12 nationalities. 50 annotators were born and living in the same country while others had moved to different ones. The vast majority of our annotators were residing in the UK (32), South Africa (9), and Ireland (7). The ratio for holding a Bachelor/Master/PhD as the highest degree was: 12.5%/76.6%/10.9%. Only 7 annotators were non-native English speakers while the other 58 were native speakers. 56.7% of the annotators self-identified as female and 43.3% as male.

3.4 Dataset Splits

We split the 10,972 validated data points into train/dev/test sets in two different ways. The stats of the two splits are shown in Table 2. In the following, we explain how they are created. **Random split:** We split the dataset randomly into train/dev/test with a ratio of 70/10/20. **Concept zero-shot split:** We create another concept zero-shot split where train/dev/test have no overlapping concepts. I.e., if “dog” appears in the train set, then

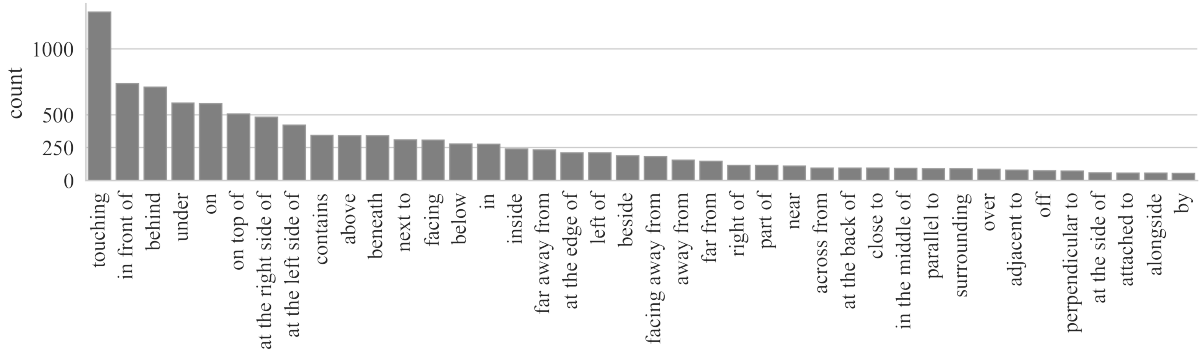


Figure 5: Relation distribution of the final dataset (sorted by frequency). Top 40 most frequent relations are included. It is clear that the relations follow a long-tailed distribution.



Figure 6: Concept distribution. Only concepts with > 100 frequencies are included.

split	train	dev	test	total
<i>random</i>	7,680	1,097	2,195	10,972
<i>zero-shot</i>	4,713	231	616	5,560

Table 2: Statistics of the *random* & *zero-shot* splits.

it does not appear in dev or test sets. This is done by randomly grouping concepts into three sets with a ratio of 50/20/30 of all concepts. This reduces the dataset size, since data points involving concepts from different parts of the train/dev/test split must be filtered out. The concept zero-shot split is a more challenging setup since the model has to learn concepts and the relations in a compositional way instead of remembering the co-occurrence statistics of the two.

3.5 Human Ceiling and Agreement

We randomly sample 500 data points from the final random split test set of the dataset for computing human ceiling and inter-annotator agreement. We hide the labels of the 500 examples and two additional annotators are asked to label `True/False` for them. On average, the two annotators achieve an accuracy of 95.4% on the VSR task. We further compute the Fleiss’ kappa among the original an-

notation and the predictions of the two human. The Fleiss’ kappa score is 0.895, indicating near-perfect agreement according to Landis and Koch (1977).

4 Dataset Analysis

In this section we compute some basic statistics of our collected data (§4.1), analyse where human annotators have agreed/disagreed (§4.2), and present a case study on reference frames (§4.3).

4.1 Basic Statistics of VSR

After the first phase of contrastive template-based caption generation (§3.1), we collected 12,809 raw data points. In the phase of the second round validation (§3.2), we collected 39,507 validation labels. Every data point received at least 3 validation labels. In 69.1% of the data points, all validators agree with the original label. 85.6% of the data points have at least $\frac{2}{3}$ annotators agreeing with the original label. We use $\frac{2}{3}$ as the threshold and exclude all instances with lower validation agreement. After excluding other instances, 10,972 data points remained and are used as our final dataset.

Here we provide basic statistics of the two components in the VSR captions: the concepts and the relations. Fig. 5 demonstrates the relation distribution. “touching” is most frequently used by